
UTILIZING SMALL QUANTUM COMPUTERS FOR MACHINE LEARNING AND GROUND-STATE ENERGY APPROXIMATION

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ABSTRACT

Limited qubit counts restrict the system sizes accessible to variational quantum algorithms. We develop a hybrid quantum-classical algorithm in which an n -qubit variational model is evaluated using quantum registers no larger than a chosen subsystem. Independent parameterized circuits prepare the subsystem states. Correlations between subsystems are represented by a structured global operation with trainable parameters, but this operation is not executed on quantum hardware. Instead, it is transferred to the observable, and the resulting Pauli expansion is evaluated as a sum of products of subsystem expectation values. Since the reconstructed expectation value depends on both the subsystem circuits and the global operation, their parameters can be optimized jointly. For variational ground-state estimation, the global operation is unitary, which ensures that the construction represents a normalized quantum state and preserves the variational interpretation of the energy. For supervised quantum machine learning, a normalized global-state interpretation is not required. Global observables may then be selected freely based on the structure of the data, gradient information, feature relevance, prior knowledge, or measurement constraints. We derive bounds for the number of local measurements, the growth of the transformed observable, the classical recombination cost, and the suppression of gradients. These bounds identify conditions under which reducing the quantum register size does not introduce an exponential measurement or post-processing overhead. We demonstrate the framework numerically using the transverse-field Ising model and handwritten-digit classification. For Ising systems containing four to sixteen qubits, the optimized energies have relative errors of order 10^{-3} . For binary classification of handwritten digits 3 and 6, a full-basis and reduced-basis model achieve test accuracies of 97.8% and 100%, respectively, on the test data split. A gradient-variance experiment with fixed eight-qubit subsystems and up to 64 modeled qubits shows no clear exponential decay over the tested range.

1 Introduction

Variational quantum algorithms combine parameterized quantum circuits with classical optimization and provide a promising framework for near-term quantum simulation and quantum machine learning [1, 2]. Their practical use on noisy intermediate-scale quantum devices is constrained by limited qubit counts, finite coherence times, imperfect gates, restricted connectivity, and measurement noise [3, 4]. Among these constraints, the number of available qubits imposes a direct limitation: a computation requiring a larger register than the available device cannot be executed without changing how the problem is represented or evaluated.

This limitation arises naturally in both application areas considered in this work. In variational ground-state estimation, the required register size is determined by the qubit representation of the Hamiltonian, so larger physical systems and more detailed representations generally require more qubits. In quantum machine learning, the register size depends on the data encoding and circuit architecture. For example, encodings that associate individual features with individual qubits require a number of qubits that grows linearly with the input dimension [5]. Consequently, both detailed physical models and high-dimensional datasets can exceed the capacity of available quantum hardware.

Several methods address this problem by replacing one large quantum computation with multiple smaller computations and classical post-processing. Circuit cutting divides a large circuit into fragments and reconstructs properties of the

original computation from measurements of the individual fragments [6, 7, 8]. Its sampling and reconstruction costs can, however, grow rapidly with the number and structure of the cuts, so its usefulness depends on whether the circuit can be partitioned with limited communication between fragments. Other approaches exploit the structure of variational many-body problems. Deep VQE first solves subsystem problems and uses selected subsystem states to construct a reduced basis for an effective global Hamiltonian, which is subsequently solved using a variational quantum algorithm [9]. This reduces the number of qubits required in each calculation, but introduces separate stages for subsystem optimization, matrix-element estimation, and reduced-space optimization. Its accuracy also depends on whether the selected subsystem states form a sufficiently expressive global basis. Entanglement forging instead represents a bipartite state through a restricted combination of product-state components [10]. Although this reduces the required quantum register, its measurement cost depends on the number of retained components, while truncation restricts the class of states that can be represented accurately.

Subsystem methods have also been proposed for quantum machine learning. Similar in spirit to circuit cutting, high-dimensional models can be decomposed into contributions evaluated on smaller subsystem registers, but with only selected terms retained rather than the full circuit being reconstructed [11]. Such truncations may still produce useful prediction functions because supervised learning does not require the exact reconstruction of a physical quantum state. The same freedom is less natural in VQE, where an arbitrarily truncated and reweighted expression does not generally correspond to the expectation value of the Hamiltonian with respect to a normalized variational state. Another register-reduction strategy is classical splitting, which replaces a single large parameterized circuit with several independent smaller circuits and constructs the model as a linear combination of their outputs [12]. This approach can reduce hardware requirements and improve local gradient behavior, but it cannot directly represent entanglement between the chosen subsystems.

In this work, we introduce a subsystem-based framework formulated directly in terms of the expectation values required by a variational objective. The qubits are partitioned into smaller subsystems, each represented by an independent parameterized quantum circuit, while correlations across subsystems are incorporated through an additional structured global operation. Rather than implementing this operation on the full quantum register, we transfer its action to the measured observables, allowing global expectation values to be reconstructed from measurements performed independently on the subsystem circuits. Importantly, the parameters of both the local subsystem circuits and the global correlation operation can be optimized jointly within a single variational procedure. We develop this construction for two settings: variational quantum eigensolvers and supervised quantum machine learning.

For VQE, the global correlation operation is chosen to be unitary, ensuring that the construction defines a normalized variational state. Its action is transferred to the Hamiltonian by applying unitary conjugation to each Hamiltonian term, after which the energy can be evaluated from expectation values of the transformed terms measured on the subsystem circuits. The principal limitation is that this transformation can substantially increase the number of Hamiltonian terms. Global operations containing many noncommuting components, or acting nonlocally across many qubits, may cause individual terms to expand into numerous contributions, with exponential growth possible in the worst case. The global operation must therefore capture relevant correlations without introducing measurement and classical-processing costs that outweigh the benefits of using smaller quantum registers.

For quantum machine learning, the model need not be interpreted as the expectation value of an observable with respect to a normalized global quantum state. Each subsystem circuit processes a subset of the input and generates local quantum features through measured expectation values, while products of these features represent correlations between subsystems. In contrast to the VQE setting, the global observables included in the learning model may be selected freely according to prior knowledge of the data structure, gradient information, or measurement cost. The resulting architecture is a hybrid model in which local quantum circuits perform feature extraction and a configurable classical layer selects and combines cross-subsystem correlations.

Reducing circuit width alone does not guarantee an efficient algorithm, because the complexity of the global model may reappear through additional measurements, classical recombination, or difficult optimization. We therefore analyze the growth of the transformed Hamiltonian, the measurement and classical recombination costs, and the scaling of gradients. We demonstrate the framework numerically through ground-state energy estimation for the transverse-field Ising model and supervised classification of handwritten digits. We also study the gradient variance as the total modeled system size increases while the individual subsystem size remains fixed.

The observable transformation used here is related to a method developed later under the name operator backpropagation [13]. Operator backpropagation transfers a later part of a quantum circuit to the measured observable to reduce the circuit depth executed on quantum hardware. The present work applies the same underlying Heisenberg-picture identity under a different resource constraint, in which the transferred operation represents correlations between independently prepared subsystems and aims to reduce the number of qubits required for each quantum computation. Operator backpropagation has also subsequently been combined with circuit cutting [14].

The remainder of the paper is organized as follows. Section 2 introduces the expectation-value decomposition. Section 3 develops the VQE construction and its implementation, and Section 4 presents the supervised-learning model. Section 5 summarizes the relevant scaling and trainability conditions, with detailed derivations provided in Appendix A. Section 6 presents the numerical demonstrations, followed by a discussion of limitations and possible extensions.

2 Subsystem-decomposed expectation values

2.1 Variational state and exact decomposition

Let an n -qubit register be partitioned into S disjoint subsystems $\mathcal{S}_1, \dots, \mathcal{S}_S$. Subsystem s contains n_s qubits, with $\sum_s n_s = n$, and is prepared as

$$|\phi_s(\boldsymbol{\theta}_s)\rangle = U_s(\boldsymbol{\theta}_s)|0\rangle_s. \quad (1)$$

The product of the locally prepared states is

$$|\Phi(\boldsymbol{\theta})\rangle = \bigotimes_{s=1}^S |\phi_s(\boldsymbol{\theta}_s)\rangle. \quad (2)$$

A full-register unitary $V(\mathbf{c})$ defines the variational state

$$|\Psi(\boldsymbol{\theta}, \mathbf{c})\rangle = V(\mathbf{c}) |\Phi(\boldsymbol{\theta})\rangle. \quad (3)$$

The state in Eq. (3) may be entangled across the subsystem partition. The purpose of the decomposition is to evaluate selected expectation values on this state without preparing the complete n -qubit state on the available quantum processor.

For an observable O , define the transformed observable

$$\tilde{O}(\mathbf{c}) = V^\dagger(\mathbf{c}) O V(\mathbf{c}). \quad (4)$$

Expanding it in Pauli strings gives

$$\tilde{O}(\mathbf{c}) = \sum_{\ell=1}^{L_O} d_\ell(\mathbf{c}) P_\ell, \quad P_\ell = \bigotimes_{s=1}^S P_\ell^{(s)}, \quad (5)$$

where $P_\ell^{(s)}$ is the restriction of P_ℓ to subsystem s . Since $|\Phi\rangle$ is a product state,

$$F(\boldsymbol{\theta}, \mathbf{c}) = \langle \Psi | O | \Psi \rangle \quad (6)$$

$$= \sum_{\ell=1}^{L_O} d_\ell(\mathbf{c}) \prod_{s=1}^S B_\ell^{(s)}(\boldsymbol{\theta}_s), \quad (7)$$

$$B_\ell^{(s)}(\boldsymbol{\theta}_s) = \langle 0_s | U_s^\dagger(\boldsymbol{\theta}_s) P_\ell^{(s)} U_s(\boldsymbol{\theta}_s) | 0_s \rangle. \quad (8)$$

Equation (7) shows that the full expectation value can be evaluated from products of expectation values measured on the individual subsystems, which are then combined classically.

2.2 Structured global unitary and observable propagation

For the VQE construction, the global unitary is chosen as an ordered product of exponentiated Pauli strings,

$$V(\mathbf{c}) = \prod_{j=1}^R V_j(c_j), \quad V_j(c_j) = e^{ic_j G_j} = \cos(c_j) \mathbb{I} + i \sin(c_j) G_j, \quad (9)$$

where each generator G_j is a Hermitian Pauli string. Since $G_j^2 = \mathbb{I}$, every factor is unitary. The generators may act across subsystem boundaries and therefore enlarge the variational family beyond product states.

For any Pauli string P ,

$$V_j^\dagger P V_j = \begin{cases} P, & [G_j, P] = 0, \\ \cos(2c_j)P + i \sin(2c_j) P G_j, & \{G_j, P\} = 0. \end{cases} \quad (10)$$

Thus the transformed observable can be obtained using only Pauli multiplication and commutation tests. No dense $2^n \times 2^n$ matrix representation of V is required.

A simple example illustrates the role of these global rotations. Let qubit a belong to one subsystem and qubit b to another, and choose $G = Z_a Y_b$. Since G anticommutes with X_a ,

$$e^{-icG} X_a e^{icG} = \cos(2c) X_a + \sin(2c) Y_a Y_b. \quad (11)$$

The second term encodes the effect of the cross-subsystem rotation and is evaluated from the product of the local Y_a and Y_b expectation values.

For an observable $O = \sum_m h_m M_m$, each initial Pauli string is propagated through the ordered factors of V ,

$$O_m^{(0)} = M_m, \quad O_m^{(j)} = V_j^\dagger O_m^{(j-1)} V_j, \quad O_m^{(R)} = V^\dagger M_m V. \quad (12)$$

At every step, Eq. (10) is applied to the current Pauli strings, and equal strings are collected. The resulting table records the coefficient of each term and its restriction to every subsystem,

$$\left\{ d_\ell(c), P_\ell^{(1)}, \dots, P_\ell^{(S)} \right\}_{\ell=1}^{L_O}. \quad (13)$$

The Pauli strings and their subsystem restrictions can be determined before optimization. During optimization, the trigonometric coefficients and the subsystem expectation values are updated.

Only distinct local Pauli strings appearing in Eq. (13) have to be measured and local expectation values can be reused in every global product in which they occur.

The generators in V can be chosen to limit the growth of the transformed observable. Branching occurs only when a generator anticommutes with a Pauli string in the current expansion. Each observable term may undergo branching under several rotations. If the observable initially contains polynomially many Pauli terms and each term undergoes branching under at most $\mathcal{O}(\log n)$ rotations, the transformed observable contains at most polynomially many Pauli strings. For local observables, choosing short generators near subsystem boundaries can introduce inter-subsystem correlations while leaving most observable terms unchanged. For example, a generator $X_i X_{i+1}$ located at a boundary commutes with all terms having disjoint support. The branching mechanism and the resulting term-growth bound are derived in Appendix A.3.

2.3 Gradients

For a parameter $\theta_i^{(q)}$ that appears only in subsystem q , differentiation of Eq. (7) gives

$$\frac{\partial F}{\partial \theta_i^{(q)}} = \sum_\ell d_\ell \frac{\partial B_\ell^{(q)}}{\partial \theta_i^{(q)}} \prod_{s \neq q} B_\ell^{(s)}. \quad (14)$$

Only the circuit on subsystem q is differentiated. For many commonly used parameterized quantum gates, including single-parameter Pauli rotations, the derivative of the local expectation value can be evaluated exactly using the parameter-shift rule [15].

For a global angle c_j , the local expectation values are unchanged and

$$\frac{\partial F}{\partial c_j} = \sum_\ell \frac{\partial d_\ell(c)}{\partial c_j} \prod_s B_\ell^{(s)}. \quad (15)$$

The coefficient derivatives follow analytically from the sine and cosine factors generated during observable propagation.

3 Variational ground-state estimation

3.1 Variational energy on subsystem circuits

Let the qubit Hamiltonian be

$$H = \sum_{m=1}^{L_H} h_m M_m, \quad (16)$$

where M_m are Pauli strings. The variational state is

$$|\Psi(\theta, c)\rangle = V(c) \left(\bigotimes_{s=1}^S U_s(\theta_s) \right) |0\rangle. \quad (17)$$

Because all factors are unitary, Eq. (17) is normalized, and its energy

$$E(\boldsymbol{\theta}, \mathbf{c}) = \langle \Psi(\boldsymbol{\theta}, \mathbf{c}) | H | \Psi(\boldsymbol{\theta}, \mathbf{c}) \rangle \quad (18)$$

is a variational upper bound to the ground-state energy.

The full-register circuit in Eq. (17) is not executed. Instead, each Hamiltonian term is transformed classically,

$$V^\dagger M_m V = \sum_{r=1}^{L_m} a_{mr}(\mathbf{c}) P_{mr}, \quad P_{mr} = \bigotimes_s P_{mr}^{(s)}, \quad (19)$$

and the energy is evaluated as

$$E(\boldsymbol{\theta}, \mathbf{c}) = \sum_{m=1}^{L_H} h_m \sum_{r=1}^{L_m} a_{mr}(\mathbf{c}) \prod_{s=1}^S \langle 0_s | U_s^\dagger P_{mr}^{(s)} U_s | 0_s \rangle. \quad (20)$$

Equation (20) is the quantity supplied to the classical optimizer. If all terms in the transformed Hamiltonian are retained, the energy is exact for the chosen ansatz in Eq. (17).

3.2 Constructing the VQE model

A practical implementation begins with the Hamiltonian and the available register size.

1. **Partition the qubits.** Divide the qubits into subsystems that fit on the available device. For a local Hamiltonian, it is suggested to keep strongly coupled qubits in the same subsystem and to place subsystem boundaries where only a limited number of Hamiltonian terms cross.
2. **Choose the subsystem ansatzes.** Assign a parameterized circuit $U_s(\boldsymbol{\theta}_s)$ to each subsystem. These circuits describe all correlations within a subsystem and are the only state-preparation circuits executed on quantum hardware.
3. **Choose the global Pauli rotations.** Select Pauli strings G_j for Eq. (9). Generators with support on more than one subsystem introduce inter-subsystem correlations. For local models, physically motivated choices are strings concentrated near partition boundaries and around Hamiltonian couplings that the product ansatz cannot represent. Their order is part of the ansatz because successive conjugations need not commute.
4. **Build the transformed Hamiltonian and measurement table.** Propagate every M_m according to Eq. (12), collect equal Pauli strings, factor each result across the partition, and deduplicate the local strings that must be measured.
5. **Measure, recombine, and optimize.** Execute the subsystem circuits, insert the measured expectation values into Eq. (20), and optimize both $\boldsymbol{\theta}$ and $\mathbf{c}$. Local parameters can be differentiated by parameter shift, while derivatives with respect to $\mathbf{c}$ are obtained from the classical coefficients as in Eq. (15).

This workflow separates the quantum and classical tasks. The quantum processor prepares and measures subsystem states. The classical processor stores the transformed Hamiltonian, evaluates products of subsystem expectation values, differentiates the coefficient functions, and updates all variational parameters.

The generator choice controls the central trade-off. Too few or poorly chosen cross-subsystem rotations leave the ansatz close to a product state. An unnecessarily large or highly nonlocal generator set can make $V^\dagger H V$ expensive to evaluate. In the numerical Ising example, the generators are chosen as short strings along and across subsystem boundaries, so the added correlations and the observable growth remain localized.

4 Supervised quantum learning

For an input vector $\mathbf{x}$, divide the features into subsets $\mathbf{x}_s$ and prepare

$$|\phi_s(\mathbf{x}_s, \boldsymbol{\theta}_s)\rangle = U_s(\mathbf{x}_s, \boldsymbol{\theta}_s) | 0_s \rangle. \quad (21)$$

A supervised model can be defined as

$$f(\mathbf{x}; \boldsymbol{\theta}, \mathbf{d}) = \sum_{p \in \mathcal{T}} d_p \prod_{s=1}^S \langle 0_s | U_s^\dagger(\mathbf{x}_s, \boldsymbol{\theta}_s) W_s^p U_s(\mathbf{x}_s, \boldsymbol{\theta}_s) | 0_s \rangle, \quad (22)$$

where $\mathcal{T}$ is a selected set of products of local Pauli strings. The quantum circuits provide local expectation-value features, and the classical layer combines products of those features.

The role of the coefficients differs from the VQE setting. In VQE, the coefficients are functions of a unitary conjugation and the complete transformed Hamiltonian is retained. For supervised learning, the output only has to define a useful prediction function. The set $\mathcal{T}$ may therefore be restricted and the coefficients d_p trained directly. This freedom controls the size of the model and can also act as a form of regularization.

The derivatives are

$$\frac{\partial f}{\partial d_q} = \prod_s B_q^{(s)}, \quad \frac{\partial f}{\partial \theta_i^{(q)}} = \sum_p d_p \frac{\partial B_p^{(q)}}{\partial \theta_i^{(q)}} \prod_{s \neq q} B_p^{(s)}. \quad (23)$$

As in the VQE construction, a circuit parameter affects only one subsystem circuit, so its derivative can be evaluated with the parameter-shift rule [15]. Possible feature-selection strategies include restricting the Pauli support, choosing a prescribed local alphabet, or retaining features with large expectation values or gradients. The numerical experiment uses a fixed pairwise feature structure between neighboring image columns.

5 Computational scaling and trainability

Reducing the size of the quantum computations does not establish an efficient algorithm itself. A reduction in quantum resources must not be replaced by an exponential measurement or classical-processing overhead. The purpose of this section is therefore to identify worst-case mechanisms through which such exponential costs may arise, together with sufficient conditions that prevent them. These conditions are worst-case conditions, but they need not be required in every implementation. The VQE and quantum machine-learning settings differ in how the relevant costs arise, while the trainability analysis applies to both. Detailed derivations are given in Appendix A. Here we summarize the resulting bounds.

5.1 Local measurement cost

The first consideration is the number of distinct local expectation values that may need to be estimated. An n_s -qubit subsystem has a complete Pauli alphabet containing 4^{n_s} observables. The worst case arises when the transformed observable contains the full local Pauli alphabet, either because all local Pauli strings are included explicitly or because the choice of V causes extensive operator growth. The total number of distinct local observables is then bounded by

$$N_{\text{local}} \leq \sum_{s=1}^S 4^{n_s}. \quad (24)$$

Choosing subsystem sizes that scale logarithmically with the total number of qubits,

$$n_s = \mathcal{O}(\log n), \quad (25)$$

is sufficient to ensure that this complete alphabet grows only polynomially with the system size. For equally sized subsystems with

$$n_s = \beta \log_2 n, \quad (26)$$

Appendix A.2 gives

$$\sum_{s=1}^S 4^{n_s} = \frac{n^{2\beta+1}}{\beta \log_2 n}. \quad (27)$$

Logarithmic subsystem width is not necessary when only a restricted set of local Pauli observables is required. If subsystem s requires $M_s \leq 4^{n_s}$ distinct local observables, then the relevant measurement count is instead

$$N_{\text{local}} = \sum_{s=1}^S M_s. \quad (28)$$

Larger subsystems can therefore remain compatible with polynomial measurement cost whenever the transformed Hamiltonian or selected feature model contains only polynomially many distinct local Pauli factors. Equations (24) and (27) should consequently be interpreted as conservative bounds for the case in which the complete local Pauli alphabet may be needed.

5.2 Classical recombination

The dominant classical cost depends on the application.

For VQE, the observables are fixed by the Hamiltonian. The relevant cost therefore comes from propagating each Hamiltonian term through the global unitary,

$$H \longrightarrow V^\dagger H V.$$

Since deleting terms from the transformed Hamiltonian changes the variational objective, the energy of the normalized state in Eq. (17) is obtained only when the complete transformed Hamiltonian is retained. Consequently, the important scaling question is how rapidly the Pauli expansion grows under successive conjugations.

Appendix A.3 shows that if a Hamiltonian term encounters k_m branching Pauli rotations during its propagation, the resulting expansion contains at most 2^{k_m} Pauli strings. For a Hamiltonian containing L_H Pauli terms, the transformed Hamiltonian therefore contains at most

$$L_H 2^k, \quad (29)$$

terms before identical Pauli strings are collected, where $k = \max_m k_m$. Thus, a polynomial-size Hamiltonian remains polynomial provided each Hamiltonian term branches only

$$k = \mathcal{O}(\log n) \quad (30)$$

times.

The situation differs in quantum machine learning, where the observables are selected as model features rather than being fixed by a Hamiltonian. In the worst case, one could include every Pauli string, resulting in an exponential classical recombination cost. Polynomial scaling therefore requires restricting the feature set. One possibility is to limit the Pauli support to at most p qubits, yielding

$$\sum_{j=0}^p \binom{n}{j} 3^j = \mathcal{O}(n^p) \quad (31)$$

candidate observables, as derived in Appendix A.4. Alternatively, only a selected subset of observables may be retained, for example through gradient-based feature selection or other importance criteria. Unlike the VQE setting, such truncation changes the expressive power of the model rather than approximating a fixed physical observable.

5.3 Trainability

Finally, reducing the number of qubits is useful only if the resulting variational model retains resolvable gradients. For a parameter $\theta_i^{(q)}$ belonging to subsystem q , the gradient can be written as a sum of Pauli-product contributions,

$$\frac{\partial F}{\partial \theta_i^{(q)}} = \sum_{\ell} X_{\ell}, \quad X_{\ell} = d_{\ell} R_q^{(\ell)} \prod_{s \neq q} B_s^{(\ell)}, \quad (32)$$

where

$$R_q^{(\ell)} = \frac{\partial B_q^{(\ell)}}{\partial \theta_i^{(q)}} \quad (33)$$

is the derivative generated by the circuit on subsystem q .

Appendix A.5 analyzes the second moment of each contribution. Under independent initialization of the subsystem circuits and of the coefficient parameters,

$$\mathbb{E}[|X_{\ell}|^2] = \mathbb{E}[|d_{\ell}|^2] \mathbb{E}\left[|R_q^{(\ell)}|^2\right] \prod_{s \neq q} \mathbb{E}\left[|B_s^{(\ell)}|^2\right]. \quad (34)$$

This separates three possible sources of suppression: the coefficient of the Pauli-product term, the derivative produced within the differentiated subsystem, and the expectation values obtained from the remaining subsystems, respectively.

For sufficiently expressive random circuits, standard barren-plateau results show that gradient moments can decrease exponentially with the number of qubits acted on by the circuit [16, 17]. In this pessimistic regime, we write the local derivative second moment as

$$\mathbb{E}\left[|R_q^{(\ell)}|^2\right] = \Theta(2^{-\gamma n_q}), \quad (35)$$

where $\gamma > 0$ depends on the circuit and observable. Choosing

$$n_q = \beta \log_2 n \quad (36)$$

therefore gives

$$2^{-\gamma n_q} = n^{-\gamma\beta}, \quad (37)$$

so that an exponential dependence on the local register size becomes an inverse-polynomial dependence on the total problem size. This is the same basic trainability mechanism used in classical splitting, where each parameter acts only within a logarithmically sized subcircuit [12].

A second source of suppression arises from the undifferentiated subsystems. If the Pauli string of term ℓ acts nontrivially on K_ℓ such subsystems of equal width $n_s = \beta \log_2 n$, then the unitary-2-design second moment gives

$$\prod_{s \neq q} \mathbb{E} \left[\left| B_s^{(\ell)} \right|^2 \right] = (n^\beta + 1)^{-K_\ell}. \quad (38)$$

Identity factors equal one and produce no suppression. Consequently, if K_ℓ is bounded independently of n , the product contributes only an inverse-polynomial dependence on the total system size.

For VQE, Appendix A.5 additionally shows that the second moment of each coefficient of the Pauli-product terms is inverse polynomial when it contains at most $\mathcal{O}(\log n)$ relevant factors. For quantum machine learning, the classical coefficient initialization can be chosen to have constant or inverse-polynomial variance.

It follows that, under the assumption that the subsystem circuits reproduce the second moments of unitary 2-designs, logarithmic subsystem width, bounded subsystem support, and inverse-polynomial coefficient moments ensure that each Pauli-product term that depends on the differentiated parameter provides a gradient contribution whose scale is not exponentially small in the total number of qubits.

These are sufficient worst-case conditions. Subsystem circuits that remain far from unitary 2-designs, including sufficiently shallow circuits with local cost functions, are expected to allow larger subsystems while retaining inverse-polynomial gradient moments [17].

6 Numerical demonstrations

The numerical studies illustrate how the construction is utilized for a physical variational energy, a supervised model, and a gradient-scaling test. All results were obtained in simulation and should be interpreted as proof-of-principle demonstrations of the method and its setup.

6.1 Transverse-field Ising model

We consider the one-dimensional transverse-field Ising Hamiltonian with periodic boundary conditions,

$$H = -J \sum_{i=1}^n X_i X_{i+1} - h \sum_{i=1}^n Z_i, \quad X_{n+1} = X_1. \quad (39)$$

In the numerical calculations, we set $J = 1$ and $h = 0.1$. The tested system sizes were $n = 4, 6, 8, 10, 12$, and 16 . Qubits were divided into equal-sized subsystems. Each subsystem used two layers of R_y rotations separated by a nearest-neighbor CNOT chain, as shown in Fig. 1.

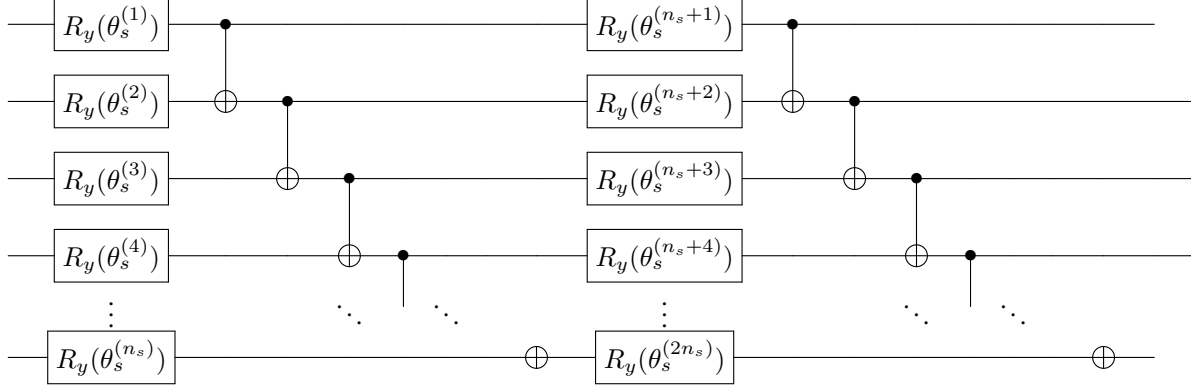


Figure 1: Subsystem ansatz used for the Ising-model experiment. Each subsystem receives a layer of trainable R_y rotations, a nearest-neighbor CNOT chain, and a second rotation-entangling layer. Only this subsystem-sized circuit is executed on quantum hardware.

The global unitary was constructed from Pauli rotations whose generators act near and across subsystem boundaries. Table 1 lists the generators used in the simulations. The choice follows the construction in Section 3.2: each subsystem circuit captures correlations within its block, while short ZY or YZ strings extend the variational ansatz across the boundaries. For example, in the eight-qubit calculation the strings Z_3Y_4 , Z_4Y_5 , and Z_5Y_6 couple the region around the boundary between qubits 4 and 5. The analogous strings near qubits 8 and 1 account for the periodic boundary. The full transformed Hamiltonian was generated from these rotations and retained in the energy evaluation.

Table 1: Subsystem partition and Pauli generators used in the Ising-model demonstration.

n	S	Generator set $\{G_j\}$
4	2	$\{Z_2Y_3, Y_1Z_4\}$
6	2	$\{Z_3Y_4, Y_1Z_6\}$
8	2	$\{Z_3Y_4, Z_4Y_5, Z_5Y_6, Z_7Y_8, Y_1Z_8, Z_1Y_2\}$
10	2	$\{Z_4Y_5, Z_5Y_6, Z_6Y_7, Z_9Y_{10}, Y_1Z_{10}, Z_1Y_2\}$
12	3	$\{Z_3Y_4, Z_4Y_5, Z_5Y_6, Z_7Y_8, Z_8Y_9, Z_9Y_{10}, Z_{11}Y_{12}, Y_1Z_{12}, Z_1Y_2\}$
16	4	$\{Z_3Y_4, Z_4Y_5, Z_5Y_6, Z_7Y_8, Z_8Y_9, Z_9Y_{10}, Z_{11}Y_{12}, Z_{12}Y_{13}, Z_{13}Y_{14}, Z_{15}Y_{16}, Y_1Z_{16}, Z_1Y_2\}$

The subsystem parameters and global-rotation angles were randomly initialized and jointly optimized with Adam [18] for 200 steps at learning rate 0.1. Figure 2 compares the resulting energies with exact diagonalization.

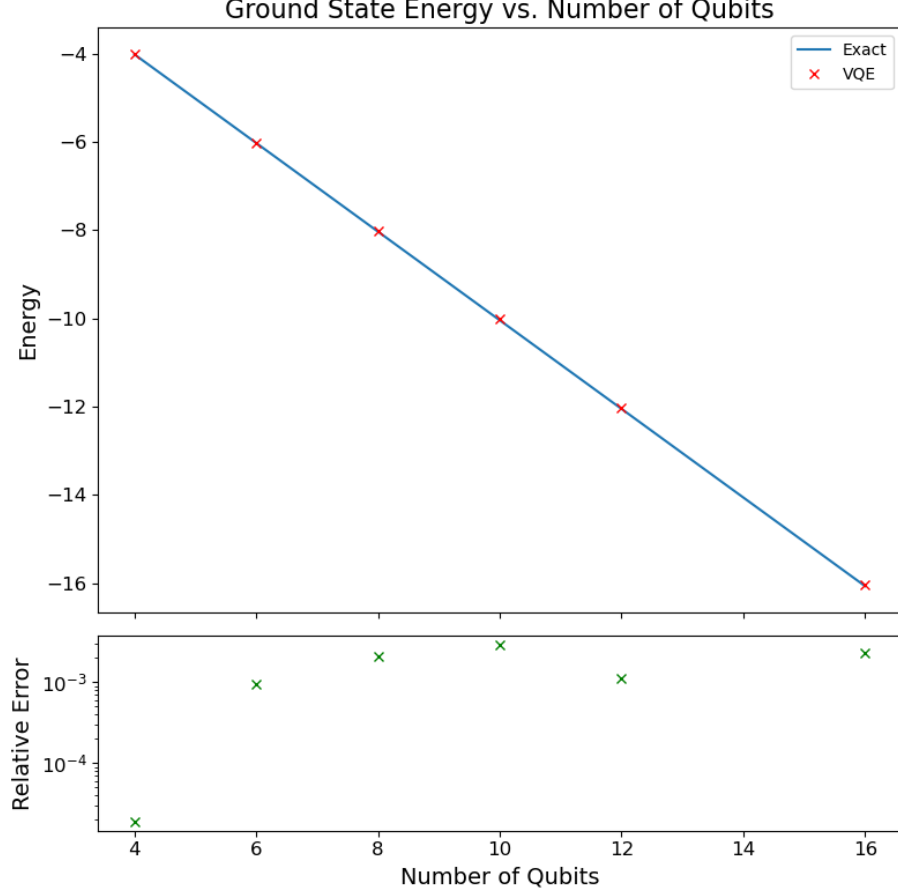


Figure 2: Ground-state energies obtained with the decomposed variational ansatz and exact diagonalization for the transverse-field Ising model. The lower panel reports the relative error. Across the tested sizes, the observed errors were of order 10^{-3} .

For these instances, the selected boundary generators improved the variational description while keeping the transformed Hamiltonian small enough to evaluate explicitly.

6.2 Handwritten-digit classification

The machine-learning demonstration used the 8×8 handwritten-digit data set and restricted the task to binary classification of digits 3 and 6 [19]. The 64 pixel values were divided by image column into eight subsystems of eight features. Pixel values were scaled to $[0, \pi]$ and encoded as R_y angles. The subsystem model used data re-uploading [20], as shown in Figs. 3 and 4.

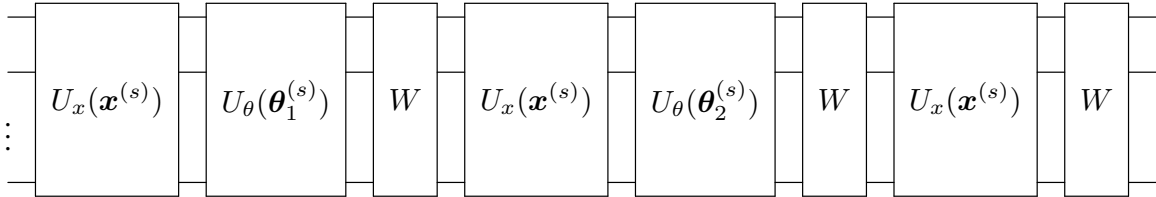


Figure 3: Eight-qubit subsystem circuit used for one image column. The feature map is re-uploaded between trainable rotation and entangling layers.

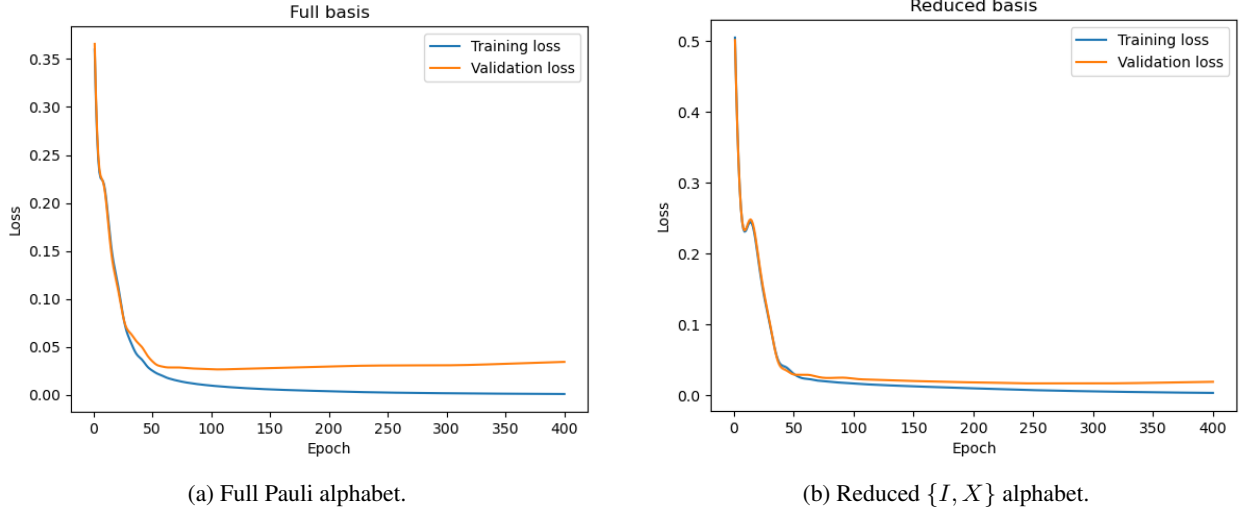


Figure 5: Training and validation losses. In both cases, validation-based model selection was used after the validation curve began to increase relative to the training curve.

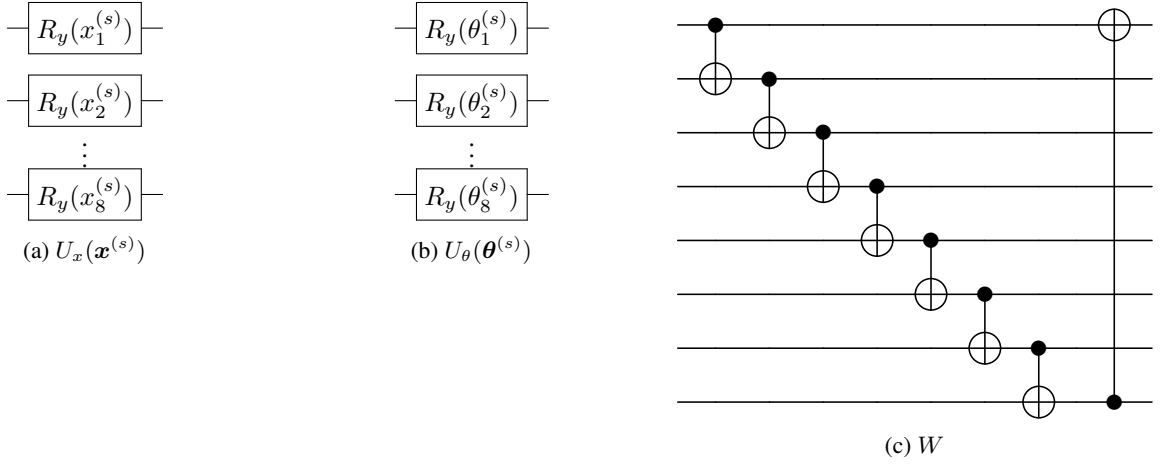


Figure 4: Components of the QML subsystem circuit: feature encoding, trainable rotations, and a ring of CNOT gates.

The observable structure connected neighboring image columns. For $s = 1, \dots, 7$,

$$D^{(s,s+1)} = \begin{cases} \sum_{a,b \in \mathcal{B}} d_{ab}^{(s)} \sigma_{1,a}^{(s)} \sigma_{1,b}^{(s+1)}, & s \text{ odd}, \\ \sum_{a,b \in \mathcal{B}} d_{ab}^{(s)} \sigma_{8,a}^{(s)} \sigma_{8,b}^{(s+1)}, & s \text{ even}, \end{cases} \quad (40)$$

where $\mathcal{B}$ was either the full single-qubit Pauli alphabet $\{I, X, Y, Z\}$ or the reduced alphabet $\{I, X\}$. The model contained 128 circuit parameters. The full-basis model added $7 \times 4^2 = 112$ classical coefficients, while the reduced-basis model added $7 \times 2^2 = 28$. Evaluation requires 56 distinct local expectation values per sample for the full-basis model and 28 for the reduced-basis model.

The 364 samples were divided into 182 training, 91 validation, and 91 test examples. Both models used mean-squared error and Adam for 400 epochs at learning rate 0.1. The final parameters were selected at the epoch with the smallest validation loss.

Table 2: Accuracy and mean-squared error for the full-basis (FB) and reduced-basis (RB) models.

Split	Accuracy (FB/RB)	MSE (FB/RB)
Training	100.0/100.0%	0.0090/0.0076
Validation	98.9/100.0%	0.0265/0.0171
Test	97.8/100.0%	0.0304/0.0250

The reduced-basis model achieved the best result in this single comparison, including 100% test accuracy. The smaller feature set may have acted as regularization, but one training run per model is insufficient to establish a statistically reliable advantage.

6.3 Gradient-variance experiment

To probe the effect of increasing total system size while keeping local circuit size fixed, the experiment began with two eight-qubit subsystems and added eight-qubit subsystems up to 64 total qubits. The observable contained all two-subsystem Pauli- X products. For each size, 20 random parameter initializations were used, and the gradient with respect to the first parameter of the first subsystem was sampled.

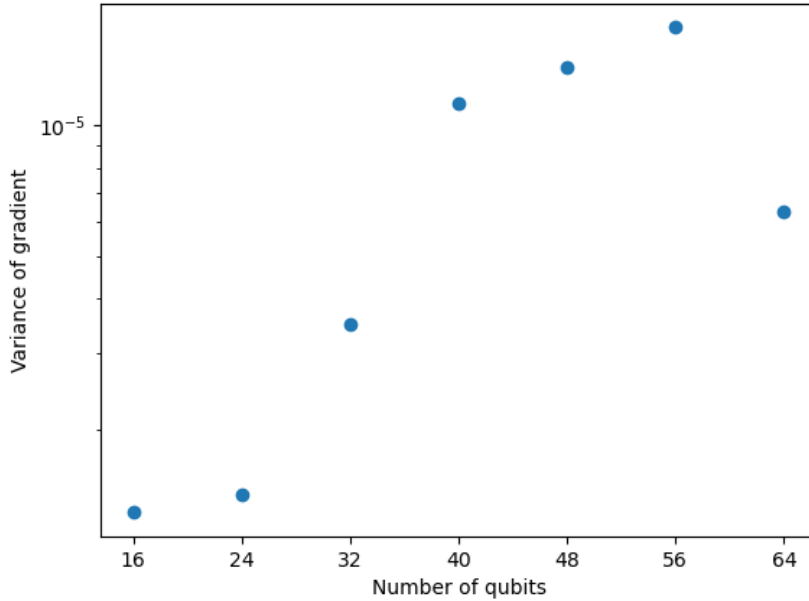


Figure 6: Empirical gradient variance as the number of eight-qubit subsystems increases. For the tested two-subsystem Pauli- X feature set, no clear exponential decay with total qubit count is observed.

The data are consistent with the support-based scaling argument. The differentiated circuit remains eight qubits wide as the number of subsystems increases, and every measured term acts nontrivially on only two subsystems. This numerical observation is specific to the chosen model and initialization sample and is not a general proof that subsystem-decomposed objectives avoid barren plateaus.

7 Discussion and limitations

The proposed method replaces a full-register circuit with a structured variational ansatz that can be evaluated through experiments on smaller subsystems. Its practical value depends on the balance between subsystem size and the complexity of the transformed observable. Increasing the subsystem size raises the quantum hardware requirements, although it also allows a larger fraction of the correlations to be represented within each local circuit. Adding more

global Pauli rotations can improve the representation of correlations between subsystems, but it can also increase the number of terms in the transformed Hamiltonian.

In VQE applications, the global generators should be chosen in relation to the structure of the Hamiltonian. The one-dimensional Ising model considered here is well suited to this construction because its interactions are local and the generators can be placed near subsystem boundaries. Hamiltonians with long-range interactions may be less favorable. The same difficulty can arise when the selected generators spread initially local Pauli terms across many qubits. In such cases, the number of measurements and the amount of classical processing may grow substantially. The method does not eliminate these costs. Instead, it exposes them through the transformed observable, which can be analyzed before the subsystem experiments are performed.

The numerical simulations were carried out using exact expectation values. On quantum hardware, finite sampling and device noise will affect both the estimated subsystem quantities and the products formed from them. These errors may also alter the optimization trajectory. Their influence was not investigated in the present study. A hardware implementation would therefore require a measurement strategy that distributes shots among the necessary observables and an analysis of how statistical errors propagate through the classical recombination procedure.

The supervised-learning model also involves an important design choice. The Pauli-product feature set can be adapted to the task, but its selection directly affects the size and expressive capacity of the model. The handwritten-digit experiment compared two fixed feature alphabets using one train-validation-test split. The reduced alphabet gave the strongest result in that comparison. Repeated training runs, alternative data splits, and additional baseline models would be needed to determine whether this behavior persists more broadly.

The trainability analysis examines the scaling of the individual terms contributing to the gradient of each quantum circuit parameter. It does not account for cancellations that may occur when these terms are summed to obtain the complete gradient. Such cancellations could suppress the total gradient even when the individual contributions remain well scaled.

8 Conclusion

We introduced a method for evaluating selected expectation values of a large variational quantum model using quantum circuits whose widths are limited to individual subsystems. Independent local circuits prepare the subsystem states, while a structured unitary represents correlations between subsystems. The action of this unitary is transferred to the observable rather than executed on a full quantum register. The resulting Pauli expansion can then be evaluated by measuring local expectation values and combining their products classically.

For variational ground-state estimation, the construction defines a normalized correlated ansatz. The energy is evaluated by transforming each Hamiltonian term, identifying the distinct local Pauli factors, measuring the corresponding subsystem expectation values, and recombining the results. The local circuit parameters and the global Pauli-rotation angles can then be optimized jointly. When the complete transformed Hamiltonian is retained, this procedure evaluates the energy of the chosen ansatz without requiring execution of the corresponding full-register circuit.

For supervised learning, the same local expectation values can be used as quantum features. Products of these features represent correlations between subsystems, while trainable classical coefficients determine how they contribute to the model output. Unlike the VQE construction, the feature set does not need to reproduce a fixed physical observable. It can therefore be restricted or adapted according to the structure of the learning task and the available measurement resources.

The scaling analysis identifies the conditions under which the reduction in circuit width is not offset by excessive measurement, recombination, or gradient costs. Under these conditions, logarithmic subsystem width keeps the local measurement cost polynomial in the total system size and ensures that gradient suppression associated with the local circuit width is at most inverse polynomial under the assumptions of the analysis. Structured global rotations limit the growth of the transformed Hamiltonian, and restricting the number of subsystems on which each term acts reduces additional suppression of gradient contributions.

The numerical experiments illustrate the method for ground-state energy estimation in the transverse-field Ising model and for binary classification of handwritten digits. They also provide an example in which the gradient variance does not show clear exponential decay over the tested system sizes. These results serve as demonstrations of the construction rather than general performance guarantees. Future studies should examine a wider range of Hamiltonians, circuit architectures, feature-selection strategies, and hardware conditions. Such studies would clarify the practical balance between ansatz expressivity, measurement cost, optimization behavior, and statistical error.

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A Scaling derivations

This appendix gives the derivations underlying the scaling statements in Section 5. The purpose of these bounds is not to predict the cost of every practical instance. Rather, they identify mechanisms through which reducing the quantum-register size could introduce an exponential measurement, classical-processing, or optimization overhead.

The relevant mechanisms differ between the two applications considered in the main text. In VQE, the observable is fixed by the Hamiltonian, and the central classical cost is the growth of the Pauli expansion under $H \mapsto V^\dagger H V$. In

quantum machine learning, the observables are chosen as model features, and the corresponding cost is determined by the number of products retained in the model. The trainability discussion applies to both settings.

A.1 Exact factorization over subsystem states

We first restate the factorization that underlies the construction. Let the state prepared before the conceptual application of the global unitary be

$$\rho_\Phi = \bigotimes_{s=1}^S \rho_s, \quad (41)$$

where ρ_s is the state prepared on subsystem s . For pure subsystem states,

$$\rho_s = U_s |0\rangle\langle 0| U_s^\dagger. \quad (42)$$

Suppose that the transformed observable has the Pauli expansion

$$\tilde{O} = V^\dagger O V = \sum_\ell d_\ell \bigotimes_{s=1}^S P_\ell^{(s)}. \quad (43)$$

Using the tensor-product structure of both ρ_Φ and each Pauli term,

$$\begin{aligned} \text{Tr}(\rho_\Phi \tilde{O}) &= \sum_\ell d_\ell \text{Tr} \left[\left(\bigotimes_{s=1}^S \rho_s \right) \left(\bigotimes_{s=1}^S P_\ell^{(s)} \right) \right] \\ &= \sum_\ell d_\ell \prod_{s=1}^S \text{Tr}(\rho_s P_\ell^{(s)}). \end{aligned} \quad (44)$$

This identity is exact when the complete Pauli expansion of $\tilde{O}$ is retained. The resource questions considered below therefore concern the number of local expectation values and classical terms required to evaluate this expansion.

A.2 Worst-case local measurement cost

The first possible source of overhead is the number of distinct local Pauli expectation values that may have to be estimated. This quantity depends on the subsystem widths.

An n_s -qubit subsystem has the Pauli basis

$$\mathcal{P}_{n_s} = \{I, X, Y, Z\}^{\otimes n_s}, \quad (45)$$

which contains 4^{n_s} strings, including the identity. The total number of distinct local Pauli strings across all subsystems is therefore bounded by

$$N_{\text{local}} \leq \sum_{s=1}^S 4^{n_s}. \quad (46)$$

For equally sized subsystems containing m qubits, the number of subsystems is $S = n/m$, and Eq. (46) becomes

$$N_{\text{local}} \leq \frac{n}{m} 4^m. \quad (47)$$

If the subsystem width is chosen as

$$m = \beta \log_2 n, \quad (48)$$

where $\beta > 0$ is independent of n , then

$$\begin{aligned}
N_{\text{local}} &\leq \frac{n}{\beta \log_2 n} 4^{\beta \log_2 n} \\
&= \frac{n}{\beta \log_2 n} 2^{2\beta \log_2 n} \\
&= \frac{n^{2\beta+1}}{\beta \log_2 n}.
\end{aligned} \tag{49}$$

Thus, logarithmic subsystem width is sufficient to make the complete local Pauli alphabet polynomial in the total number of qubits.

Equation (49) establishes that logarithmic subsystem width is sufficient to prevent the complete local Pauli alphabet from growing exponentially with the total system size. This is, however, a worst-case guarantee rather than a necessary condition. In a concrete calculation, one measures only the local Pauli strings that actually appear in the transformed Hamiltonian or in the selected feature set, and the same local expectation value may be reused across several global terms. Consequently, the relevant measurement cost is determined by the number of distinct local Pauli factors that occur, which can remain polynomial even for subsystems larger than $\mathcal{O}(\log n)$. The logarithmic-width condition is therefore required only when one allows for the possibility that the full local Pauli alphabet must be measured.

A.3 Growth of the transformed Hamiltonian in VQE

For VQE, the observable is fixed by the Hamiltonian. The energy of the normalized variational state in Eq. (17) is

$$E = \langle \Phi | V^\dagger H V | \Phi \rangle. \tag{50}$$

Discarding terms from $V^\dagger H V$ changes the objective and therefore no longer gives the exact variational energy of the chosen ansatz. The relevant classical scaling question is how many Pauli terms can be generated when the Hamiltonian is propagated through the global unitary.

We write the global unitary as

$$V = \prod_{j=1}^R V_j, \quad V_j = e^{ic_j G_j}, \tag{51}$$

where each generator G_j is a Hermitian Pauli string satisfying $G_j^2 = I$. The generators may have different supports and may act within one subsystem or across several subsystems.

Consider first the conjugation of a Pauli string P by a single rotation. Since

$$e^{\pm icG} = \cos(c)I \pm i \sin(c)G, \tag{52}$$

we obtain

$$\begin{aligned}
e^{-icG} P e^{icG} &= (\cos(c)I - i \sin(c)G) P (\cos(c)I + i \sin(c)G) \\
&= \cos^2(c)P + i \cos(c) \sin(c)(PG - GP) + \sin^2(c)GPG.
\end{aligned} \tag{53}$$

If G and P commute, then $GPG = P$ and $PG - GP = 0$, so

$$e^{-icG} P e^{icG} = P. \tag{54}$$

If they anticommute, then $GPG = -P$ and $GP = -PG$, which gives

$$e^{-icG} P e^{icG} = \cos(2c)P + i \sin(2c)PG. \tag{55}$$

Because iPG is, up to an overall sign, another Hermitian Pauli string, one rotation maps a Pauli string either to itself or to a sum of at most two Pauli strings. Growth of the expansion therefore occurs only when a current Pauli string anticommutes with the next generator.

To track this growth, let

$$M_m^{(0)} = M_m, \quad M_m^{(j)} = V_j^\dagger M_m^{(j-1)} V_j \quad (56)$$

denote the expansion obtained after propagating the Hamiltonian term M_m through the first j rotations. Let $N_m^{(j-1)}$ be the number of distinct Pauli strings present immediately before the j th conjugation, after equal strings have been collected. Suppose that $b_m^{(j)}$ of these strings anticommute with G_j . The remaining $N_m^{(j-1)} - b_m^{(j)}$ strings commute with G_j and remain unchanged, whereas each anticommuting string produces at most two terms. It follows that

$$\begin{aligned} N_m^{(j)} &\leq (N_m^{(j-1)} - b_m^{(j)}) + 2b_m^{(j)} \\ &= N_m^{(j-1)} + b_m^{(j)} \leq 2N_m^{(j-1)}. \end{aligned} \quad (57)$$

A rotation is therefore said to produce branching during the propagation of M_m when $b_m^{(j)} > 0$, that is, when at least one Pauli string in the current expansion anticommutes with G_j . A rotation for which $b_m^{(j)} = 0$ leaves the number of terms unchanged, while a branching rotation can increase it by at most a factor of two.

Let k_m denote the number of rotations that produce branching while propagating M_m through the complete sequence. Since the initial expansion contains one term, $N_m^{(0)} = 1$, repeated application of Eq. (57) gives

$$N_m^{(R)} \leq 2^{k_m}. \quad (58)$$

This is a worst-case bound in which the number of terms doubles at every branching rotation. In practice, only some of the current strings may anticommute with a given generator. Repeated Pauli strings, cancellations between coefficients, and parameter values for which a sine or cosine factor vanishes can further reduce the final expansion.

For a Hamiltonian

$$H = \sum_{m=1}^{L_H} h_m M_m, \quad (59)$$

the number of Pauli terms in the transformed Hamiltonian is bounded, before collecting identical strings originating from different Hamiltonian terms, by

$$N_{V^\dagger H V} \leq \sum_{m=1}^{L_H} 2^{k_m} \leq L_H 2^k, \quad k = \max_m k_m. \quad (60)$$

If the original Hamiltonian contains a polynomial number of Pauli terms and

$$k = \mathcal{O}(\log n), \quad (61)$$

then the bound in Eq. (60) is polynomial in the total number of qubits.

The relevant complexity parameter is not the total number of rotations in V or the support of the individual generators. It is the number of rotations that cause branching during the propagation of each Hamiltonian term. Since each conjugation may generate new Pauli strings, branching must be determined from the complete ordered sequence of generators.

A generator may have broad support without increasing the expansion if it commutes with every Pauli string present at that stage. Conversely, several short generators may produce substantial growth if they repeatedly anticommute with the generated strings. The size of the transformed Hamiltonian must therefore be evaluated for the complete generator sequence.

The bound in Eq. (60) is a worst-case bound. Commutation relations, repeated strings, Hamiltonian symmetries, coefficient cancellations, and particular parameter values may reduce the actual expansion. Nevertheless, the transformed Hamiltonian remains polynomial in size when the original Hamiltonian contains polynomially many terms and each term undergoes branching under at most $\mathcal{O}(\log n)$ rotations.

A.4 Feature complexity in quantum machine learning

The scaling question is different in quantum machine learning because the observables are not fixed by a Hamiltonian. They are selected as features of the model. It is therefore permissible to omit terms, but doing so changes the expressive class of the model.

Suppose initially that every global Pauli string is included. Since an n -qubit register has 4^n Pauli strings, the number of possible feature terms is

$$N_{\text{all}} = 4^n. \quad (62)$$

The same result follows from the subsystem decomposition. If subsystem s has 4^{n_s} local Pauli strings and every product of local strings is included, then

$$\prod_{s=1}^S 4^{n_s} = 4^{\sum_s n_s} = 4^n. \quad (63)$$

Equations (62) and (63) show that logarithmic subsystem width alone does not make an unrestricted feature model efficient. Although the local measurement alphabet can be polynomial, the number of global products formed from those local quantities can still be exponential. A polynomially scalable model must therefore retain only a structured subset of the possible products.

One way to restrict the model is to bound the number of qubits on which a Pauli string acts nontrivially. A Pauli string with support on exactly j qubits is specified by choosing those qubits and assigning one of X , Y , or Z to each selected position. The number of strings with nonidentity support on at most p qubits is therefore

$$N_{\leq p} = \sum_{j=0}^p \binom{n}{j} 3^j. \quad (64)$$

For fixed p ,

$$\binom{n}{j} \leq \frac{n^j}{j!}. \quad (65)$$

It follows that

$$N_{\leq p} = \mathcal{O}(n^p). \quad (66)$$

The polynomial scaling in Eq. (66) follows from [21]. Thus, a model containing every Pauli string up to a fixed support size is polynomially large.

More generally, the model may select a polynomial-sized set $\mathcal{T}$ directly rather than retaining every observable satisfying a support condition. Such a set may be fixed from physical considerations, as in the neighboring-column structure used in the numerical experiment, or selected adaptively using coefficient magnitudes, gradients, or another criterion. Omitting a candidate Pauli string defines a smaller model, so feature restriction is therefore part of model design and can also act as a form of regularization.

A.5 Termwise gradient scaling and subsystem support

We now derive the trainability conditions summarized in Section 5. The gradient is evaluated as a classical sum of contributions associated with the Pauli products appearing in the transformed observable. We determine the scale of each contribution separately and derive conditions under which each retained term provides a gradient contribution that is not exponentially small in the total system size.

For a parameter $\theta_i^{(q)}$ belonging to subsystem q , Eq. (14) gives

$$\frac{\partial F}{\partial \theta_i^{(q)}} = \sum_{\ell} d_{\ell} R_q^{(\ell)} \prod_{s \neq q} B_s^{(\ell)}, \quad (67)$$

where

$$R_q^{(\ell)} = \frac{\partial}{\partial \theta_i^{(q)}} \langle 0_q | U_q^\dagger P_\ell^{(q)} U_q | 0_q \rangle \quad (68)$$

is evaluated entirely on subsystem q , and

$$B_s^{(\ell)} = \langle 0_s | U_s^\dagger P_\ell^{(s)} U_s | 0_s \rangle \quad (69)$$

is the expectation value measured on subsystem s .

Define the contribution from Pauli-product term ℓ as

$$X_\ell = d_\ell R_q^{(\ell)} \prod_{s \neq q} B_s^{(\ell)}. \quad (70)$$

The complete gradient is then

$$\frac{\partial F}{\partial \theta_i^{(q)}} = \sum_\ell X_\ell. \quad (71)$$

We study the second moment

$$\mathbb{E}[|X_\ell|^2]. \quad (72)$$

The subsystem circuits are initialized independently. The circuit on subsystem q is therefore statistically independent of those on the remaining subsystems, and quantities obtained from different subsystem circuits are mutually independent. If d_ℓ is also initialized independently of the local circuit parameters, then

$$\mathbb{E}[|X_\ell|^2] = \mathbb{E}[|d_\ell|^2] \mathbb{E}\left[|R_q^{(\ell)}|^2\right] \prod_{s \neq q} \mathbb{E}\left[|B_s^{(\ell)}|^2\right]. \quad (73)$$

If d_ℓ is deterministic, the first expectation is replaced by $|d_\ell|^2$.

Equation (73) separates the contribution into three factors,

$$\mathbb{E}[|d_\ell|^2], \quad \mathbb{E}\left[|R_q^{(\ell)}|^2\right], \quad \prod_{s \neq q} \mathbb{E}\left[|B_s^{(\ell)}|^2\right]. \quad (74)$$

The first is determined by the coefficient of the Pauli product. The second depends on the width and structure of the differentiated circuit. The third depends on the support of the Pauli product across the remaining subsystems.

A.5.1 Local derivative contribution

The derivative $R_q^{(\ell)}$ is generated entirely by the n_q qubits in subsystem q . Its scaling is therefore controlled by the local circuit width rather than directly by the total number of qubits in the complete problem.

For sufficiently expressive or highly randomized parameterized circuits, gradient variances can decrease exponentially with the number of qubits [16]. The scaling also depends strongly on circuit depth and cost-function locality [17]. To represent the pessimistic random-circuit regime, we assume that the local derivative second moment has the asymptotic scale

$$\mathbb{E}\left[|R_q^{(\ell)}|^2\right] = \mathcal{O}(2^{-\gamma n_q}), \quad (75)$$

where $\gamma > 0$ depends on the local circuit, its parameterization, and the measured observable.

Let the differentiated subsystem contain

$$n_q = \beta \log_2 n \quad (76)$$

qubits, where $\beta > 0$ is independent of n . It follows that

$$2^{-\gamma n_q} = 2^{-\gamma \beta \log_2 n} = n^{-\gamma \beta}. \quad (77)$$

Thus, the exponential dependence on the local circuit width becomes an inverse-polynomial dependence on the total system size. This is the same mechanism employed in classical splitting, where each parameter belongs to a subcircuit containing at most $\mathcal{O}(\log n)$ qubits [12].

The logarithmic-width condition follows specifically from the exponential model in Eq. (75). More generally, suppose that a circuit construction gives

$$\mathbb{E} \left[\left| R_q^{(\ell)} \right|^2 \right] = \mathcal{O}(\chi(n_q)). \quad (78)$$

The subsystem width may then be increased as long as

$$\chi(n_q) \geq \frac{1}{\text{poly}(n)}. \quad (79)$$

For example, if a structured local circuit gives

$$\chi(n_q) = n_q^{-\zeta} \quad (80)$$

for some constant $\zeta > 0$, then a width

$$n_q = \mathcal{O}(n^\alpha) \quad (81)$$

with fixed $\alpha > 0$ still gives

$$\chi(n_q) = \mathcal{O}(n^{-\alpha\zeta}), \quad (82)$$

which is inverse polynomial. Techniques that improve the local gradient scaling can therefore allow larger subsystems than the logarithmic worst-case condition. Shallow circuits and appropriately local cost functions provide examples of such improved behavior [17].

A.5.2 Contribution from nondifferentiated subsystems

The second source of suppression is the product

$$\prod_{s \neq q} B_s^{(\ell)} \quad (83)$$

in Eq. (70). Independence of the subsystem circuits gives

$$\mathbb{E} \left[\left| \prod_{s \neq q} B_s^{(\ell)} \right|^2 \right] = \prod_{s \neq q} \mathbb{E} \left[\left| B_s^{(\ell)} \right|^2 \right]. \quad (84)$$

Let

$$d_s = 2^{n_s} \quad (85)$$

be the Hilbert-space dimension of subsystem s . For a pure input state ρ_s and a unitary drawn from a unitary 2-design, the second-moment identity is [17].

$$\mathbb{E}_U \left[\text{Tr}(U \rho_s U^\dagger A) \text{Tr}(U \rho_s U^\dagger C) \right] = \frac{\text{Tr}(A) \text{Tr}(C) + \text{Tr}(AC)}{d_s(d_s + 1)}. \quad (86)$$

Setting

$$A = C = P_\ell^{(s)} \quad (87)$$

in Eq. (86) gives

$$\mathbb{E} \left[\left| B_s^{(\ell)} \right|^2 \right] = \frac{\text{Tr} \left(P_\ell^{(s)} \right)^2 + \text{Tr} \left[\left(P_\ell^{(s)} \right)^2 \right]}{d_s(d_s + 1)}. \quad (88)$$

If

$$P_\ell^{(s)} = \mathbb{I}, \quad (89)$$

then

$$B_s^{(\ell)} = 1 \quad (90)$$

for every normalized subsystem state. Consequently,

$$\mathbb{E} \left[\left| B_s^{(\ell)} \right|^2 \right] = 1. \quad (91)$$

If

$$P_\ell^{(s)} \neq \mathbb{I}, \quad (92)$$

then the Pauli string is traceless and squares to the identity:

$$\text{Tr}(P_\ell^{(s)}) = 0, \quad \text{Tr}\left[\left(P_\ell^{(s)}\right)^2\right] = d_s. \quad (93)$$

Equation (88) therefore becomes

$$\mathbb{E}\left[\left|B_s^{(\ell)}\right|^2\right] = \frac{1}{d_s + 1} = \frac{1}{2^{n_s} + 1}. \quad (94)$$

Define the set of nondifferentiated subsystems on which term ℓ acts nontrivially as

$$\mathcal{R}_\ell^{(q)} = \left\{s \neq q : P_\ell^{(s)} \neq \mathbb{I}\right\}, \quad (95)$$

and let

$$K_\ell = \left|\mathcal{R}_\ell^{(q)}\right|. \quad (96)$$

Using Eqs. (91) and (94),

$$\prod_{s \neq q} \mathbb{E}\left[\left|B_s^{(\ell)}\right|^2\right] = \prod_{s \in \mathcal{R}_\ell^{(q)}} \frac{1}{2^{n_s} + 1}. \quad (97)$$

For equally sized subsystems with

$$n_s = \beta \log_2 n, \quad (98)$$

Eq. (97) becomes

$$\prod_{s \neq q} \mathbb{E}\left[\left|B_s^{(\ell)}\right|^2\right] = \frac{1}{(n^\beta + 1)^{K_\ell}}. \quad (99)$$

If K_ℓ is bounded by a constant independent of n , equivalently $K_\ell = \mathcal{O}(1)$, then for each fixed value of K_ℓ ,

$$(n^\beta + 1)^{-K_\ell} = \mathcal{O}(n^{-\beta K_\ell}). \quad (100)$$

The product therefore gives only inverse-polynomial suppression as n grows. All identity factors on the remaining subsystems contribute exactly one.

In contrast, if K_ℓ grows extensively with the number of subsystems, the product in Eq. (99) may become superpolynomially or exponentially small. Subsystem support is therefore a separate trainability criterion from subsystem width.

Since Eq. (94) follows from the unitary-2-design assumption, the resulting logarithmic-width condition is only a sufficient worst-case condition. As for the differentiated-subsystem contribution, larger subsystems are admissible whenever the actual local second moments decay sufficiently slowly.

A.5.3 Coefficient scaling in VQE

In quantum machine learning, the coefficients d_ℓ are trainable classical parameters, and their initialization variance can be chosen to remain constant or inverse polynomial in n . In VQE, the coefficients are $h_m a_{mr}$ according to Eq. (20). The Hamiltonian coefficient h_m is fixed by the problem, while a_{mr} is generated by propagating M_m through the global Pauli rotations in V .

The rotations in V are applied sequentially to the Pauli expansion of the Hamiltonian. If a Pauli string P commutes with the next generator G_j , the corresponding term remains unchanged. If it anticommutes with G_j , then

$$V_j^\dagger P V_j = \cos(2c_j)P + i \sin(2c_j)P G_j. \quad (101)$$

The current term is replaced by two terms. One carries a cosine factor and the other carries a sine factor. Both resulting Pauli strings are then propagated through the remaining rotations. At every later rotation, each current term splits again whenever its Pauli string anticommutes with the corresponding generator.

The coefficient a_{mr} can therefore be written as

$$a_{mr}(\mathbf{c}) = \sigma_{mr} \prod_{j \in \mathcal{B}_{mr}^{(c)}} \cos(2c_j) \prod_{j \in \mathcal{B}_{mr}^{(s)}} \sin(2c_j), \quad (102)$$

where $\sigma_{mr} \in \{-1, 1\}$ contains the signs arising from Pauli multiplication. The sets $\mathcal{B}_{mr}^{(c)}$ and $\mathcal{B}_{mr}^{(s)}$ contain the branching rotations that contribute a cosine or sine factor, respectively. The total number of such factors is

$$J_{mr} = \left| \mathcal{B}_{mr}^{(c)} \right| + \left| \mathcal{B}_{mr}^{(s)} \right|. \quad (103)$$

Assume that the global angles are initialized independently according to

$$c_j \sim \text{Unif}[0, 2\pi]. \quad (104)$$

The required second moments are

$$\mathbb{E}_c[\cos^2(2c)] = \mathbb{E}_c[\sin^2(2c)] = \frac{1}{2}. \quad (105)$$

Since the factors in Eq. (102) depend on independently initialized angles,

$$\begin{aligned} \mathbb{E}_c[|a_{mr}|^2] &= \prod_{j \in \mathcal{B}_{mr}^{(c)}} \mathbb{E}_{c_j}[\cos^2(2c_j)] \prod_{j \in \mathcal{B}_{mr}^{(s)}} \mathbb{E}_{c_j}[\sin^2(2c_j)] \\ &= \left(\frac{1}{2}\right)^{J_{mr}}. \end{aligned} \quad (106)$$

Let k_m be the number of rotations that produce branching during the propagation of M_m , as defined in Appendix A.3. Every coefficient generated from M_m satisfies

$$J_{mr} \leq k_m. \quad (107)$$

Under the logarithmic branching condition

$$k_m \leq \kappa \log_2 n, \quad (108)$$

we obtain

$$\begin{aligned} \mathbb{E}_c[|a_{mr}|^2] &= \left(\frac{1}{2}\right)^{J_{mr}} \\ &\geq \left(\frac{1}{2}\right)^{k_m} \\ &\geq \left(\frac{1}{2}\right)^{\kappa \log_2 n} = n^{-\kappa}. \end{aligned} \quad (109)$$

The logarithmic branching condition from Appendix A.3 therefore ensures that each a_{mr} has an inverse-polynomial second moment.

The coefficient that enters the energy and gradient contributions is $h_m a_{mr}$. Its second moment is

$$\mathbb{E}_c[|h_m a_{mr}|^2] = |h_m|^2 \mathbb{E}_c[|a_{mr}|^2]. \quad (110)$$

Suppose that

$$|h_m|^2 = \mathcal{O}(n^{-\eta_m}) \quad (111)$$

for some constant $\eta_m \geq 0$. Equations (109) and (111) then give

$$\mathbb{E}_c[|h_m a_{mr}|^2] = \mathcal{O}(n^{-\eta_m - \kappa}). \quad (112)$$

Thus, the coefficient of each Pauli-product contribution in Eq. (20) has an inverse-polynomial second moment when the logarithmic branching condition is satisfied and the corresponding Hamiltonian coefficient decreases no faster than inverse polynomially.

A.5.4 Combined scaling of one contribution

For the general decomposition, the contribution associated with Pauli-product term ℓ is

$$X_\ell = d_\ell R_q^{(\ell)} \prod_{s \neq q} B_s^{(\ell)}. \quad (113)$$

Combining Eqs. (73), (75), and (97) gives

$$\mathbb{E}[|X_\ell|^2] = \mathcal{O} \left[\mathbb{E}(|d_\ell|^2) 2^{-\gamma n_q} \prod_{s \in \mathcal{R}_\ell^{(q)}} \frac{1}{2^{n_s} + 1} \right] \quad (114)$$

under the stated independence and local-gradient assumptions.

For equally sized logarithmic subsystems,

$$n_s = n_q = \beta \log_2 n, \quad (115)$$

this becomes

$$\mathbb{E}[|X_\ell|^2] = \mathcal{O}\left[\mathbb{E}(|d_\ell|^2) n^{-\gamma\beta} (n^\beta + 1)^{-K_\ell}\right]. \quad (116)$$

Suppose that

$$\mathbb{E}[|d_\ell|^2] = \mathcal{O}(n^{-\delta_\ell}) \quad (117)$$

for some constant $\delta_\ell \geq 0$, and that K_ℓ is bounded independently of n . Equations (116) and (100) then give

$$\mathbb{E}[|X_\ell|^2] = \mathcal{O}(n^{-\delta_\ell - \gamma\beta - \beta K_\ell}). \quad (118)$$

In the VQE setting, the coefficient d_ℓ associated with the contribution indexed by (m, r) is

$$d_{mr} = h_m a_{mr}. \quad (119)$$

Under Eqs. (108) and (111),

$$\mathbb{E}[|d_{mr}|^2] = \mathcal{O}(n^{-\eta_m - \kappa}). \quad (120)$$

The corresponding gradient contribution is

$$X_{mr} = h_m a_{mr} R_q^{(mr)} \prod_{s \neq q} B_s^{(mr)}. \quad (121)$$

For the Pauli string P_{mr} appearing in the expansion

$$V^\dagger M_m V = \sum_r a_{mr} P_{mr}, \quad (122)$$

define the set of nondifferentiated subsystems on which this conjugated Pauli string acts nontrivially as

$$\mathcal{R}_{mr}^{(q)} = \left\{ s \neq q : P_{mr}^{(s)} \neq \mathbb{I} \right\}, \quad K_{mr} = |\mathcal{R}_{mr}^{(q)}|. \quad (123)$$

For logarithmic subsystem widths and bounded K_{mr} , Eq. (118) becomes

$$\mathbb{E}[|X_{mr}|^2] = \mathcal{O}(n^{-\eta_m - \kappa - \gamma\beta - \beta K_{mr}}). \quad (124)$$

The coefficient, local-circuit, and subsystem-support factors are therefore all inverse polynomial when each Hamiltonian term undergoes branching under at most $\mathcal{O}(\log n)$ rotations, the Hamiltonian coefficients decrease no faster than inverse polynomially, the differentiated subsystem contains $\mathcal{O}(\log n)$ qubits under the pessimistic local-gradient model, and each Pauli product acts nontrivially on only a bounded number of nondifferentiated subsystems. Consequently, each Pauli-product term that depends on the differentiated parameter provides a gradient contribution whose scale is not exponentially small in the total number of qubits.

For quantum machine learning, the corresponding coefficient condition can be enforced directly by initializing the trainable classical coefficients with constant or inverse-polynomial variance.